

GRADE 9 MATHEMATICS · STATISTICS & PROBABILITY

Statistics & Data Analy

THE BIG IDEA

Correlation is not causation
your data range are far mo
predictions beyond it.

LINE OF BEST FIT — RULES

1. Drawn through the **middle** of the data

ysis

n — and predictions **inside**
re reliable than

2. Follows the **direction** of the trend

3. Does NOT need to touch the origin or

WORKED EXAMPLE

Study hours vs. test score, line through (1

Slope: $m = (90 - 52) / (6 - 1) = 7.6$

Equation: $y = 7.6x + 44.4$

Predict 3.5 h: 71.0% (inside range — reliable)

Predict 10 h: 120.4% — impossible, **unreliable**

⚠ WATCH OUT FOR

MISTAKE

“The line must pass through the origin”

any data point

(1,52) and (6,90)

able)

able extrapolation

FIX

The line of best fit can have any y-intercept

MISTAKE

“Strong correlation proves causation”

MISTAKE

“Extrapolation is just as reliable”



SELF-CHECK

Can you describe the correlation in a scatter
find the equation of a line of best fit from two

Correlation is one piece of evidence, not proof

FIX

It grows increasingly uncertain beyond the data range

plot using direction and strength? Can you
o points?